Single-qubit states and superposition

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video

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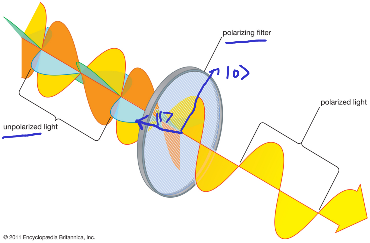
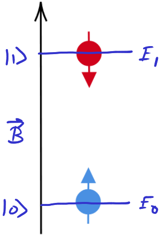
Simplest quantum system: qubit

(2 level sym) $d=2$



Polarization = qubit

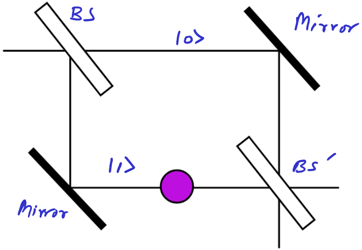


Spin

in mag field

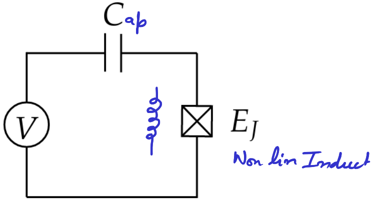
$E = -\vec{\mu}\cdot\vec{B}$

Quantum dot



Mach–Zehnder interferometer

Time bin qubit



Superconducting charge qubit

Qubit

$$\cos\Phi = 1 - \frac{\phi^2}{2!} + \frac{\phi^4}{4!}$$

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Single-qubit "mixed" state

state space is a Convex set

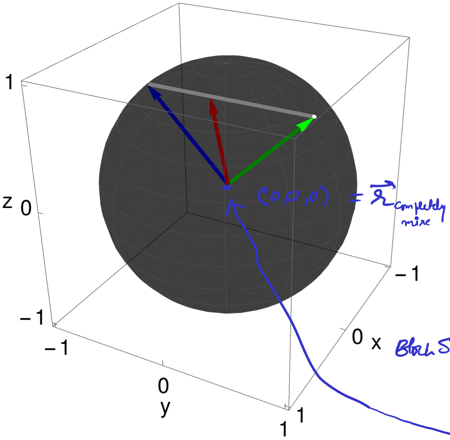
$d=2$

Convex combination

$$\rho_{\text{mix}} = w\,|\psi\rangle\langle\psi| + (1-w)|\chi\rangle\langle\chi|\,, \quad 0 < w < 1$$

$$= w\,\frac{I + \hat{r}\cdot\vec{\sigma}}{2} + (1-w)\,\frac{I + \hat{s}\cdot\vec{\sigma}}{2} \qquad |\hat{r}| = 1 = |\hat{s}|$$

$$= \frac{I + \overbrace{(w\,\hat{r} + (1-w)\,\hat{s})}^{\vec{r}_{\text{mix}}}\cdot\vec{\sigma}}{2}$$



$$\vec{r}_{\text{mix}} = w\,\hat{r} + (1-w)\,\hat{s}\,, \quad |\vec{r}_{\text{mix}}| < 1$$

Bloch Sphere = Boundary points $\leftrightarrow$ pure states

Interior points $\leftrightarrow$ mixed states

$$\rho = \frac{I}{2} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$$

Completely mixed state

Bloch Ball

$$\rho = \frac{1}{d} I$$

Single-qubit "pure" state

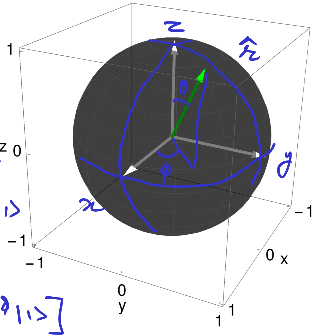
$|\psi\rangle = \underbrace{\left[|a_0\rangle\langle a_0| + |a_1\rangle\langle a_1|\right]}_{I} |\psi\rangle$

Pure state vector (ket):

$$|\psi\rangle = |a_0\rangle \underbrace{\langle a_0|\psi\rangle}_{\alpha_0} + |a_1\rangle \underbrace{\langle a_1|\psi\rangle}_{\alpha_1} \equiv \begin{pmatrix}\alpha_0 \\ \alpha_1\end{pmatrix} \text{ in } \mathcal{B}_a = \{|a_0\rangle, |a_1\rangle\}.$$

$p_i = |\alpha_i|^2$ prob amp

$\mathbb{C}^2 = \mathcal{H}$



- Two complex numbers α_0 and α_1 that obeys

  Norm: $\langle\psi|\psi\rangle = |\alpha_0|^2 + |\alpha_1|^2 = 1$ can be parameterized by three real numbers as

$$\alpha_0 = e^{i\gamma} \cos\frac{\theta}{2} \qquad \alpha_1 = e^{i\gamma} \sin\frac{\theta}{2} e^{i\phi}$$

$$|\psi\rangle = e^{i\gamma}\left(\cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}\, e^{i\phi}\, |1\rangle\right) = e^{i\gamma}\, |\psi'\rangle$$

global phase factor

zx plane $\phi = 0, \pi$

$|\psi\rangle = \cos\theta/2\,|0\rangle \pm \sin\theta/2\,|1\rangle$

xy plane $\theta = \pi/2$

$|\psi\rangle = \frac{1}{\sqrt{2}}\left[|0\rangle + e^{i\phi}|1\rangle\right]$

yz plane $\phi = \frac{\pi}{2}, \frac{3\pi}{2}$

$|\psi\rangle = \cos\theta/2\,|0\rangle \pm i \sin\theta/2\,|1\rangle$

- γ has no observable consequence

$\langle\psi'|\cancel{e^{-i\gamma}}\, \Pi\, \cancel{e^{i\gamma}}|\psi'\rangle$

$$\langle\Pi\rangle = \langle\psi|\Pi|\psi\rangle = \langle\psi'|\Pi|\psi'\rangle$$ Born Rule

Pure quantum state $|\psi\rangle\langle\psi| = |\psi'\rangle\langle\psi'| = \rho$

$\cancel{e^{i\gamma}}|\psi'\rangle\langle\psi'|\cancel{e^{-i\gamma}}$

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Single-qubit "pure" state

$SU(2)$ $\quad |\psi\rangle \in \mathcal{H} \equiv \mathbb{C}^2$

$SO(3)$ $\quad \hat{r} \in \mathbb{R}^3$ $\qquad (\theta, \phi) \in \mathbb{R}^2$

$$\rho = |\psi\rangle\langle\psi| \equiv \begin{pmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2}\, e^{i\phi} \end{pmatrix} \begin{pmatrix} \cos\frac{\theta}{2} & \sin\frac{\theta}{2}\, e^{-i\phi} \end{pmatrix} \quad \text{in } \mathcal{B} = \{|0\rangle, |1\rangle\}$$

$$= \begin{pmatrix} (\cos\frac{\theta}{2})^2 & \cos\frac{\theta}{2}\sin\frac{\theta}{2}\, e^{-i\phi} \\ \cos\frac{\theta}{2}\sin\frac{\theta}{2}\, e^{i\phi} & (\sin\frac{\theta}{2})^2 \end{pmatrix}$$

$i = \sqrt{-1}$

$$= \begin{pmatrix} \frac{1+\cos\theta}{2} & \frac{\sin\theta}{2} e^{-i\phi} \\ \frac{\sin\theta}{2} e^{i\phi} & \frac{1-\cos\theta}{2} \end{pmatrix}$$

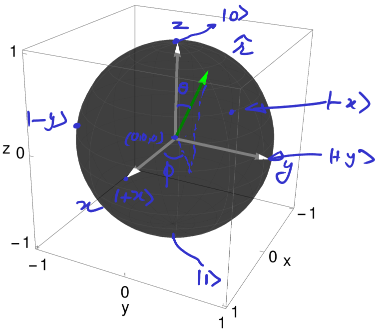
$e^{-i\phi} \to \cos\phi - i\sin\phi$

$e^{i\phi} = \cos\phi + i\sin\phi$

$$= \frac{I + \hat{r}\cdot\vec{\sigma}}{2}$$

$|\pm\rangle = |\pm x\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$

$|\pm y\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle)$



Bloch vector: $\hat{r} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta) = (x, y, z)$, $|\hat{r}| = 1$

$x = \langle X\rangle = \langle\psi|X|\psi\rangle$
$= tr(\rho X)$

Pauli vector operator: $\vec{\sigma} = (X, Y, Z)$

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$X|\pm x\rangle = \pm 1\, |\pm x\rangle$

$Y|\pm y\rangle = \pm 1\, |\pm y\rangle$

$Z|0\rangle = +1|0\rangle$

$Z|1\rangle = -1|1\rangle$

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